

dirac16complex00 in the primordial gravitational field

The semi-classical $\text{Pin}(4,4)$ spinor field: Lagrangian, field equations, non-triviality [2], energy-momentum tensor, equations of state and the field equations for $a_4[x_4]$ (Revision record)

dirac16complex project (with Claude Opus 5.5)

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Abstract

This document records, for the author's primordial gravitational field of `Revision/SPEC.md` section 1, the field theory of `dirac16complex00`: a field Φ with 16 complex commuting components that transform as a spinor of $\text{Pin}(4,4)$ (a set of 16 scalar fields that transform as a $\text{Pin}(4,4)$ spinor), treated as a classical ("semi-classical") field. It gives the Lagrangian and its coupling to gravity through the vielbein and the canonical spin connection, the exact Euler-Lagrange equations (covariant, explicit in the metric, and as 16 component equations), the non-triviality theorem [2] with its proof (the gravitational term $\gamma^\mu \Omega_\mu = 3H\gamma^{(x_8)}$ of the field equation never vanishes for $H > 0$), the self-consistency of the equations, the energy-momentum tensor as a classical bilinear, its kinetic and potential parts, the energy density, the pressures and the equations of state, and the Einstein-Lovelock field equations for $a_4(x_4)$ with `dirac16complex00` as the source. Every formula is an output of Revision code (Wolfram and an independent sympy checker for each part); each section ends with the names of the checks that verify it, and the check index collects them. Nothing is taken from the earlier stages of the repository. What the record does not establish is stated in the section "What is not claimed".

1 Scope, sources and conventions

Sources. Every formula and number below is read from these Revision outputs (all produced by Revision code, all checks passing):

part	Wolfram record	independent sympy record
gammas, $\text{Pin}(4,4)$, $\text{Spin}(4,4)$ (SPEC section 2)	Revision/algebra /reports/wolfram -algebra.json, fixture Revision/algebra/gammas. json	Revision/algebra /reports/python-algebra. json
geometry, Lagrangian, field equations, energy-momentum tensor (SPEC sections 1-4)	Revision/theory/reports/ wolfram-field-theory.json, formulas Revision/theory/ field-theory.json	Revision/theory/reports/ python-field-theory.json
field equations for $a_4(x_4)$ (SPEC section 5)	Revision/field_equations_a4/reports/wolfram-a4-report.json, formulas Revision/field_equations_a4/a4-equations.json	Revision/field_equations_a4/reports/python-a4-report.json
pairing theorems (SPEC section 9)	Revision/pairing /reports/wolfram -pairing.json, theorems Revision/pairing /pairing-theory.json	Revision/pairing /reports/python-pairing. json

The Wolfram and the sympy records are written independently (no shared code); each sympy record re-derives the statements and compares them with the Wolfram record. The theory records verify every statement for both statistics: the checks whose names end in `_C` (Wolfram) or start with

`commuting_` (sympy) are the ones for the commuting field `dirac16complex00` of this document; the Grassmann versions belong to `dirac16complex` ([Revision/docs/DIRAC16COMPLEX_FIELD_THEORY](#)).

Coordinates (the author's names). x_1, x_2, x_3 are ordinary 3-space (inflating, scale factor $e^{a_4} \sin^{1/6} z$); x_4 is the time; x_5, x_6, x_7 are the three extra times, time-like, which deflate exponentially (scale factor $e^{-a_4} \sin^{1/6} z$ with a_4 increasing); x_8 is the hidden space direction. Throughout $z = 6Hx_8 \in (0, \pi/2)$, $H > 0$ is the author's constant and $a'_4 = da_4/dx_4$. In formulas the index i runs over x_1, x_2, x_3 and t over x_5, x_6, x_7 .

The metric (SPEC section 1, exactly):

```
g = {{E^(2 a4[x4]) Sin[6 H x8]^(1/3), 0, 0, 0, 0, 0, 0, 0},
      {0, E^(2 a4[x4]) Sin[6 H x8]^(1/3), 0, 0, 0, 0, 0, 0},
      {0, 0, E^(2 a4[x4]) Sin[6 H x8]^(1/3), 0, 0, 0, 0, 0},
      {0, 0, 0, -1, 0, 0, 0, 0},
      {0, 0, 0, 0, -E^(-2 a4[x4]) Sin[6 H x8]^(1/3), 0, 0, 0},
      {0, 0, 0, 0, 0, -E^(-2 a4[x4]) Sin[6 H x8]^(1/3), 0, 0},
      {0, 0, 0, 0, 0, 0, -E^(-2 a4[x4]) Sin[6 H x8]^(1/3), 0},
      {0, 0, 0, 0, 0, 0, 0, Cot[6 H x8]^2}}
```

Signature (4,4): space-like x_1, x_2, x_3, x_8 , time-like x_4, x_5, x_6, x_7 . The volume factor is

$$\sqrt{|g|} = \cos z,$$

independent of x_4 : the inflation e^{3a_4} of 3-space is compensated by the deflation e^{-3a_4} of the extra times. $H = 0$ is not a metric of the family (a degenerate limit: $g_{11}, g_{55} \rightarrow 0$ and $g_{88} \rightarrow \infty$); every statement assumes $H > 0$ and $0 < z < \pi/2$.

Spinor notation. $\bar{\Phi} = \Phi^\dagger C$, $S = \bar{\Phi}\Phi$, $U(S)$ a real potential ($U = \frac{\lambda}{2}S^2$ where stated), $V = m + U'(S)$. Frame gammas are written $\gamma^{(x_a)}$, coordinate gammas $\gamma^\mu = e^\mu_a \gamma^{(a)}$. The theory records write Ψ for both statistics; here Φ denotes the commuting field.

Labels. "Exact" means verified by exact symbolic computation (no floating point) in the named checks; "interpretation" marks a reading that is not itself a computed statement.

statement	Wolfram checks	sympy checks
the metric, its vielbein and $\sqrt{ g } = \cos z$	<code>metric_is_the_authors,</code> <code>sqrt_det_g_is_cos_z</code>	<code>metric_from_viel</code> <code>bein_equals_SPEC,</code> <code>sqrt_det_g_equals_cos_z</code>
$H = 0$ is a degenerate limit	<code>degenerate_at_H_0</code>	

2 The field `dirac16complex00` and its symmetry group

`dirac16complex00` is a field Φ with 16 complex commuting components (16 scalar fields) that transform as a $\text{Pin}(4,4)$ spinor. It is a classical field, the analogue of Dirac's 1928 wave function; it is not quantised (SPEC section 6: only `dirac16complex`, the anticommuting field, is quantised canonically).

The gammas. The real 16×16 matrices are the author's T16, rebuilt in Revision code from the author's formulas (the tau matrices and their blocks), with the frame map $\gamma^{(x_8)} = T16[0]$, $\gamma^{(x_1 \dots x_3)} = T16[1..3]$, $\gamma^{(x_4)} = T16[4]$ and $\gamma^{(x_5 \dots x_7)} = T16[5..7]$. Verified exactly:

- $\{\gamma^{(a)}, \gamma^{(b)}\} = 2\eta^{ab}$ with $\eta = \text{diag}(+1, +1, +1, -1, -1, -1, -1, +1)$ in the order $x_1 \dots x_8$; every $\gamma^{(a)}$ is a real signed permutation matrix; the space-like ones are symmetric, the time-like ones antisymmetric.
- $C = \gamma^{(x_8)}\gamma^{(x_1)}\gamma^{(x_2)}\gamma^{(x_3)}$ is real symmetric, $C^2 = 1$, and $C\gamma^{(a)}$ is real antisymmetric for every a ; $C = \text{diag}(-\sigma, \sigma)$ with the notebook's σ .
- $\Gamma = \gamma^{(x_8)}\gamma^{(x_1)} \dots \gamma^{(x_7)} = \text{diag}(-I_8, I_8)$, $\Gamma^2 = 1$, and Γ anticommutes with every $\gamma^{(a)}$.
- $B = -iC\gamma^{(x_4)}$ is Hermitian, $B^2 = 1$, with eigenvalues $+1$ and -1 eight times each: signature $(8,8)$.
- $S^{ab} = \frac{1}{4}[\gamma^{(a)}, \gamma^{(b)}]$ satisfy the $\text{so}(4,4)$ relations; $(S^{ab})^T C + C S^{ab} = 0$, so $S = \bar{\Phi}\Phi$ is $\text{Spin}(4,4)$ -invariant; $[\Gamma, S^{ab}] = 0$.

Pin(4,4) and Spin(4,4). The 16-dimensional representation is irreducible under $\text{Pin}(4,4)$ (commutant dimension 1: the 256 Clifford products span all of $M_{16}(\mathbb{R})$). Under $\text{Spin}(4,4)$ it splits into the two chiral halves $\Gamma = -1$ and $\Gamma = +1$, each irreducible (commutant dimension 1 on each half), and the two 8-dimensional representations are inequivalent (intertwiner dimension 0; the commutant of all S^{ab} on the 16 has dimension 2, spanned by the chiral projectors). Every reflection $\gamma^{(a)}$ exchanges the halves. $\text{Pin}(4,4)$ is the double cover of $\text{O}(4,4)$, $\text{Spin}(4,4)$ of $\text{SO}(4,4)$.

Consequence used below. Since $C^2 = 1$ and $C = \text{diag}(-\sigma, \sigma)$ is not ± 1 , C has both eigenvalues $+1$ and -1 : the bilinear $S = \Phi^\dagger C \Phi$ is an indefinite Hermitian form and takes both signs.

statement	Wolfram checks (algebra)	sympy checks (algebra)
Clifford relation, reality, symmetry pattern	Clifford_relation, reality, symmetry _pattern, eta_in_author_order, coordinate_map	clifford_relation, clifford_relation_sympy, reality_signed_permutations, symmetry _pattern
C, Γ, B	C_real_symmetric, C_squared_identity, C_gamma_real_antisymmetric, Gamma_diag, B_Hermitian, B_squared_identity, B_signature_8_8	C_equals_notebook_sigma16, C_real_symmetric_involution, C_gamma_antisymmetric, chirality_diag, chirality_anticommutes, B_hermitian_involution_signature
$\text{so}(4,4)$, invariance of S	S_Lorentz_algebra, S_preserves_C, S_commutates_with_Gamma	S_lorentz_algebra, S_preserves_C_and_commutates_with_Gamma

statement	Wolfram checks (algebra)	sympy checks (algebra)
Pin(4,4) irreducible; Spin(4,4): two inequivalent irreducible halves	Pin44_irreducibl e_commutant_dim_1, Spin44_commutant _dim_2_chiral_pr ojectors, chiral_h alves_irreducible, chiral_halves_in equivalent_inter twiners_0, Spin_Pin _reflection_swap s_halves	clifford_product s_span_M16, pin_comm utant_dimension_1, even_products_sp an_M8_plus_M8, spin_com mutant_dimension_2, spin_halves_irreducible, spin_halves_ineq uivalent, reflecti ons_exchange_halves
the fixture equals the independent construction		fixture_comparis on_gammas_json, gammas_j son_equals_pytho n_construction

3 Coupling to the primordial field: vielbein and canonical spin connection

Vielbein. The diagonal vielbein $e^a{}_\mu = f_a \delta^a_\mu$ with

$$\begin{aligned} f_{x_1} = f_{x_2} = f_{x_3} &= e^{a_4} \sin^{1/6} z, & f_{x_4} &= 1, \\ f_{x_5} = f_{x_6} = f_{x_7} &= e^{-a_4} \sin^{1/6} z, & f_{x_8} &= \cot z \end{aligned}$$

reproduces the author's metric entry by entry, $g = e^T \eta e$, with inverse $e_a{}^\mu = \delta_a^\mu / f_a$ and $\sqrt{|g|} = f_1 \cdots f_8 = \cos z$.

Levi-Civita connection. The 25 independent nonzero Christoffel symbols ($m \leq n$) are, for $i = x_1, x_2, x_3$ and $t = x_5, x_6, x_7$ (no sum),

$$\begin{aligned} \Gamma^{x_i}_{x_i x_4} &= a'_4, & \Gamma^{x_i}_{x_i x_8} &= H \cot z, & \Gamma^{x_4}_{x_i x_i} &= e^{2a_4} \sin^{1/3} z a'_4, \\ \Gamma^{x_8}_{x_i x_i} &= -e^{2a_4} H \sec z \sin^{4/3} z, \\ \Gamma^{x_t}_{x_4 x_t} &= -a'_4, & \Gamma^{x_t}_{x_t x_8} &= H \cot z, & \Gamma^{x_4}_{x_t x_t} &= e^{-2a_4} \sin^{1/3} z a'_4, \\ \Gamma^{x_8}_{x_t x_t} &= e^{-2a_4} H \sec z \sin^{4/3} z, & \Gamma^{x_8}_{x_8 x_8} &= -12H \csc(12H x_8). \end{aligned}$$

Canonical spin connection. $\omega_\mu{}^a{}_b = e^a{}_\nu (\partial_\mu e_b{}^\nu + \Gamma^\nu_{\mu\lambda} e_b{}^\lambda)$ satisfies the vielbein postulate $\partial_\mu e^a{}_\nu - \Gamma^\lambda_{\mu\nu} e^a{}_\lambda + \omega_\mu{}^a{}_b e^b{}_\nu = 0$ for all 512 components, and $\omega_{\mu ab} = \eta_{ac} \omega_\mu{}^c{}_b = -\omega_{\mu ba}$. Exactly 12 components with $a < b$ are nonzero:

$$\begin{aligned} \omega_{x_i, (x_i)(x_4)} &= a'_4 e^{a_4} \sin^{1/6} z, & \omega_{x_i, (x_i)(x_8)} &= H e^{a_4} \sin^{1/6} z, \\ \omega_{x_t, (x_4)(x_t)} &= -a'_4 e^{-a_4} \sin^{1/6} z, & \omega_{x_t, (x_t)(x_8)} &= -H e^{-a_4} \sin^{1/6} z, \end{aligned}$$

and $\omega_{x_4} = \omega_{x_8} = 0$. Each component is a'_4 or H times a factor that never vanishes on the patch.

Spinor connection and covariant derivative. $\Omega_\mu = \frac{1}{2}\omega_{\mu ab}S^{ab}$:

$$\begin{aligned}\Omega_{x_i} &= \frac{1}{2}e^{a_4} \sin^{1/6} z (a'_4 \gamma^{(x_i)} \gamma^{(x_4)} + H \gamma^{(x_i)} \gamma^{(x_8)}), & i = x_1, x_2, x_3, \\ \Omega_{x_t} &= -\frac{1}{2}e^{-a_4} \sin^{1/6} z (a'_4 \gamma^{(x_4)} \gamma^{(x_t)} + H \gamma^{(x_t)} \gamma^{(x_8)}), & t = x_5, x_6, x_7, \\ \Omega_{x_4} &= \Omega_{x_8} = 0,\end{aligned}$$

$$D_\mu \Phi = \partial_\mu \Phi + \Omega_\mu \Phi, \quad D_\mu \bar{\Phi} = \partial_\mu \bar{\Phi} - \bar{\Phi} \Omega_\mu, \quad \gamma^\mu = e^\mu_a \gamma^{(a)}.$$

This is the canonical connection: it makes the gammas covariantly constant,

$$D_\mu \gamma^\nu = \partial_\mu \gamma^\nu + \Gamma^\nu_{\mu\lambda} \gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0$$

for all 64 pairs (μ, ν) . It is built from the mixed $\omega_\mu^a{}_b$ lowered with η (SPEC section 3), not from a contraction of the mixed components without the metric factor.

Curvature. $R^\mu{}_\nu$ is diagonal with

$$R^{x_i}{}_{x_i} = a''_4 - 6H^2, \quad R^{x_4}{}_{x_4} = 6(a'_4)^2, \quad R^{x_t}{}_{x_t} = -a''_4 - 6H^2, \quad R^{x_8}{}_{x_8} = -6H^2,$$

$$R = 6((a'_4)^2 - 7H^2).$$

Since $R^{x_8}{}_{x_8} = -6H^2 < 0$ for every $H > 0$ and every function a_4 , the metric is curved for every member of the family, a_4 constant included; all Riemann components vanish only in the formal limit $a'_4 = a''_4 = 0$, $H = 0$ (flat 4+4), which is outside the family.

statement	Wolfram checks (theory)	sympy checks (theory)
vielbein and its inverse	metric_is_the_authors, vielbein_inverse	metric_from_viel bein_equals_SPEC
25 Christoffel symbols	christoffel_count	christoffel_symm etric_metric_compatible
vielbein postulate, antisymmetry, the 12 components	vielbein_postulate, omega_antisymmetric, omega_components	vielbein_postulate, spin_connection_ antisymmetric
Ω_μ ; $\Omega_{x_4} = \Omega_{x_8} = 0$	Omega_components	Omega_x4_and_Ome ga_x8_vanish
covariant constancy of the gammas	gamma_covariantl y_constant	covariant_consta ncy_D_mu_gamma_nu
Ricci tensor and scalar; curved for every $H > 0$	ricci_mixed_components, ricci_scalar, never_fl at_for_H_positive	curvature_nonzer o_flat_only_formally

4 The Lagrangian

The Lagrangian density (SPEC section 3; the same form as for dirac16complex, only the statistics differ):

$$\mathcal{L} = \sqrt{|g|} \left[\frac{1}{2} (\bar{\Phi} \gamma^\mu D_\mu \Phi - (D_\mu \bar{\Phi}) \gamma^\mu \Phi) - m S - U(S) \right], \quad S = \bar{\Phi} \Phi = \Phi^\dagger C \Phi,$$

with an explicit mass term linear in m and quadratic in the spinor, and $U(S) = \frac{\lambda}{2} S^2$ where stated (a general real U elsewhere). Proved exactly for the commuting field:

1. $\mathcal{L}$ is real, $\mathcal{L}^* = \mathcal{L}$: C real symmetric, $C\gamma^{(a)}$ real antisymmetric and $(S^{ab})^T C + C S^{ab} = 0$ make the symmetrised kinetic term and $\bar{\Phi}\Phi$ real without a factor i (the sympy record checks it on 408 monomials, with the control that the unsymmetrised form is not real off shell).
2. $\mathcal{L}$ differs from the unsymmetrised form by a total divergence: $\mathcal{L} = \sqrt{|g|} [\bar{\Phi} \gamma^\mu D_\mu \Phi - m S - U] - \frac{1}{2} \partial_\mu (\sqrt{|g|} \bar{\Phi} \gamma^\mu \Phi)$.
3. For this diagonal metric $\{\gamma^\mu, \Omega_\mu\} = 0$ for each μ separately, so $\bar{\Phi} \{\gamma^\mu, \Omega_\mu\} \Phi = 0$ and the spin connection drops out of $\mathcal{L}$ itself (next display).

$$\mathcal{L} = \cos z \left[\frac{1}{2} \sum_a \frac{1}{f_a} (\bar{\Phi} \gamma^{(a)} \partial_a \Phi - \partial_a \bar{\Phi} \gamma^{(a)} \Phi) - m S - U(S) \right].$$

The canonical spin connection nevertheless reaches the field equations, through the identity

$$\partial_\mu (\sqrt{|g|} \gamma^\mu) = 2 \sqrt{|g|} \gamma^\mu \Omega_\mu$$

in the variation of the derivative terms (section 6).

Negative control (why the Dirac-type form is used). The notebook's real Majorana-type Lagrangian $L_g = \sqrt{|g|} \Theta^T C \gamma^\mu D_\mu \Theta$ with 16 real anticommuting components is a total derivative,

$$L_g = \partial_\mu (\frac{1}{2} \sqrt{|g|} \Theta^T C \gamma^\mu \Theta),$$

and has no field equations (all Euler-Lagrange expressions vanish identically). For real commuting components the same L_g is not a total derivative: it gives the massless expression $2\sqrt{|g|} (C \gamma^\mu D_\mu \Phi)_A$, while $\Phi^T C \gamma^\mu \Phi = 0$ identically. The Dirac-type $\mathcal{L}$ with $\bar{\Phi} = \Phi^\dagger C$ is used for both fields: it is real, carries the mass term, and gives the same field equation for both statistics.

statement	Wolfram checks (theory)	sympy checks (theory)
$\mathcal{L}$ real	L_real_C	commuting_lagran gian_real, commutin g_controls_not_vacuous
total-divergence relation	L_total_divergen ce_to_unsymmetrised_C	commuting_total_ divergence_relation
spin connection drops out of $\mathcal{L}$, enters through the divergence identity	L_spin_connectio n_drops_out_C, gamma0me ga_divergence_form	anticommutator_g amma_Omega_vanishes, divergence_of_sq rtg_gamma

statement	Wolfram checks (theory)	sympy checks (theory)
negative control:	Majorana_Lg_tota	negative_control
Majorana-type L_g	l_derivative_grassmann,	_majorana_grassm
	Majorana_Lg_comm	ann_total_derivative,
	uting_control	negative_control
		_majorana_commut
		ing_contrast

5 The exact field equations

Covariant form. The Euler-Lagrange equations of $\mathcal{L}$ for the commuting field (variation with respect to Φ^* and to Φ) are

$$\gamma^\mu D_\mu \Phi = (m + U'(S)) \Phi, \quad (D_\mu \bar{\Phi}) \gamma^\mu = -(m + U'(S)) \bar{\Phi},$$

and the second is the Dirac conjugate of the first. For $U = \frac{\lambda}{2} S^2$: $\gamma^\mu D_\mu \Phi = (m + \lambda S) \Phi$. (The sympy derivation includes the control that the same comparison with $m \rightarrow -m$ fails.)

Explicit form in the primordial field ($z = 6Hx_8$):

$$\begin{aligned} & e^{-a_4} \sin^{-1/6} z (\gamma^{(x_1)} \partial_1 + \gamma^{(x_2)} \partial_2 + \gamma^{(x_3)} \partial_3) \Phi + \gamma^{(x_4)} \partial_4 \Phi \\ & + e^{a_4} \sin^{-1/6} z (\gamma^{(x_5)} \partial_5 + \gamma^{(x_6)} \partial_6 + \gamma^{(x_7)} \partial_7) \Phi \\ & + \tan z \gamma^{(x_8)} \partial_8 \Phi + 3H \gamma^{(x_8)} \Phi = (m + U'(S)) \Phi, \end{aligned}$$

and the adjoint equation

$$\begin{aligned} & e^{-a_4} \sin^{-1/6} z (\partial_1 \bar{\Phi} \gamma^{(x_1)} + \partial_2 \bar{\Phi} \gamma^{(x_2)} + \partial_3 \bar{\Phi} \gamma^{(x_3)} + \partial_4 \bar{\Phi} \gamma^{(x_4)} \\ & + e^{a_4} \sin^{-1/6} z (\partial_5 \bar{\Phi} \gamma^{(x_5)} + \partial_6 \bar{\Phi} \gamma^{(x_6)} + \partial_7 \bar{\Phi} \gamma^{(x_7)} \\ & + \tan z \partial_8 \bar{\Phi} \gamma^{(x_8)} + 3H \bar{\Phi} \gamma^{(x_8)} = -(m + U'(S)) \bar{\Phi}. \end{aligned}$$

The derivative coefficients are $1/f_a$: the inverse 3-space scale factor $e^{-a_4} \sin^{-1/6} z$, the inverse extra-time scale factor $e^{a_4} \sin^{-1/6} z$ (it grows as the extra times deflate) and $\tan z$ for the hidden direction; the last term on the left is the spin-connection term.

Evolution form. Since $(\gamma^{(x_4)})^2 = -1$, the field equation is equivalent to the first-order evolution equation

$$\partial_4 \Phi = -\gamma^{(x_4)} \left[(m + U'(S)) \Phi - \sum_{a \neq x_4} \frac{1}{f_a} \gamma^{(a)} \partial_a \Phi - 3H \gamma^{(x_8)} \Phi \right];$$

the slices $x_4 = \text{const}$ are non-characteristic ($g^{44} = -1 \neq 0$).

Chiral block form. With $\Phi = (\varphi_-, \varphi_+)$ ($\Gamma = -1$ on rows 1-8, $\Gamma = +1$ on rows 9-16) every gamma is block off-diagonal, with 8×8 blocks $\bar{\tau}_a$ (upper right) and τ_a (lower left) listed in `Revision/theory/field-theory.json` (`field_equation_blocks`), and the field equation is the pair

$$\sum_a \frac{1}{f_a} \bar{\tau}_a \partial_a \varphi_+ + 3H \bar{\tau}_{x_8} \varphi_+ = V \varphi_-, \quad \sum_a \frac{1}{f_a} \tau_a \partial_a \varphi_- + 3H \tau_{x_8} \varphi_- = V \varphi_+.$$

The mass term and the gravitational term $3H$ both couple the two chiral halves.

The 16 component equations. With $u = e^{-a_4} \sin^{-1/6} z$, $v = e^{a_4} \sin^{-1/6} z$, $t = \tan z$, $V = m + U'(S)$ and Φ_A the components in the row order of `Revision/algebra/gammas.json`, the equation $(\gamma^\mu D_\mu \Phi)_A = V \Phi_A$ reads (generated from `field_equation_components` of `Revision/theory/field-theory.json`, which the sympy record compares term by term):

$$\begin{aligned} (1) \quad & -\partial_4 \Phi_{14} + u(-\partial_1 \Phi_{16} + \partial_2 \Phi_{15} - \partial_3 \Phi_{14}) \\ & + v(\partial_5 \Phi_{15} + \partial_6 \Phi_{16} + \partial_7 \Phi_9) + (t \partial_8 + 3H) \Phi_9 = V \Phi_1 \\ (2) \quad & \partial_4 \Phi_{13} + u(-\partial_1 \Phi_{15} - \partial_2 \Phi_{16} + \partial_3 \Phi_{13}) \\ & + v(\partial_5 \Phi_{16} - \partial_6 \Phi_{15} + \partial_7 \Phi_{10}) + (t \partial_8 + 3H) \Phi_{10} = V \Phi_2 \\ (3) \quad & \partial_4 \Phi_{16} + u(\partial_1 \Phi_{14} - \partial_2 \Phi_{13} - \partial_3 \Phi_{16}) \\ & + v(-\partial_5 \Phi_{13} + \partial_6 \Phi_{14} + \partial_7 \Phi_{11}) + (t \partial_8 + 3H) \Phi_{11} = V \Phi_3 \\ (4) \quad & -\partial_4 \Phi_{15} + u(\partial_1 \Phi_{13} + \partial_2 \Phi_{14} + \partial_3 \Phi_{15}) \\ & + v(-\partial_5 \Phi_{14} - \partial_6 \Phi_{13} + \partial_7 \Phi_{12}) + (t \partial_8 + 3H) \Phi_{12} = V \Phi_4 \\ (5) \quad & \partial_4 \Phi_{10} + u(-\partial_1 \Phi_{12} + \partial_2 \Phi_{11} - \partial_3 \Phi_{10}) \\ & + v(-\partial_5 \Phi_{11} - \partial_6 \Phi_{12} - \partial_7 \Phi_{13}) + (t \partial_8 + 3H) \Phi_{13} = V \Phi_5 \\ (6) \quad & -\partial_4 \Phi_9 + u(-\partial_1 \Phi_{11} - \partial_2 \Phi_{12} + \partial_3 \Phi_9) \\ & + v(-\partial_5 \Phi_{12} + \partial_6 \Phi_{11} - \partial_7 \Phi_{14}) + (t \partial_8 + 3H) \Phi_{14} = V \Phi_6 \\ (7) \quad & -\partial_4 \Phi_{12} + u(\partial_1 \Phi_{10} - \partial_2 \Phi_9 - \partial_3 \Phi_{12}) \\ & + v(\partial_5 \Phi_9 - \partial_6 \Phi_{10} - \partial_7 \Phi_{15}) + (t \partial_8 + 3H) \Phi_{15} = V \Phi_7 \\ (8) \quad & \partial_4 \Phi_{11} + u(\partial_1 \Phi_9 + \partial_2 \Phi_{10} + \partial_3 \Phi_{11}) \\ & + v(\partial_5 \Phi_{10} + \partial_6 \Phi_9 - \partial_7 \Phi_{16}) + (t \partial_8 + 3H) \Phi_{16} = V \Phi_8 \\ (9) \quad & \partial_4 \Phi_6 + u(\partial_1 \Phi_8 - \partial_2 \Phi_7 + \partial_3 \Phi_6) \\ & + v(-\partial_5 \Phi_7 - \partial_6 \Phi_8 - \partial_7 \Phi_1) + (t \partial_8 + 3H) \Phi_1 = V \Phi_9 \\ (10) \quad & -\partial_4 \Phi_5 + u(\partial_1 \Phi_7 + \partial_2 \Phi_8 - \partial_3 \Phi_5) \\ & + v(-\partial_5 \Phi_8 + \partial_6 \Phi_7 - \partial_7 \Phi_2) + (t \partial_8 + 3H) \Phi_2 = V \Phi_{10} \\ (11) \quad & -\partial_4 \Phi_8 + u(-\partial_1 \Phi_6 + \partial_2 \Phi_5 + \partial_3 \Phi_8) \\ & + v(\partial_5 \Phi_5 - \partial_6 \Phi_6 - \partial_7 \Phi_3) + (t \partial_8 + 3H) \Phi_3 = V \Phi_{11} \\ (12) \quad & \partial_4 \Phi_7 + u(-\partial_1 \Phi_5 - \partial_2 \Phi_6 - \partial_3 \Phi_7) \\ & + v(\partial_5 \Phi_6 + \partial_6 \Phi_5 - \partial_7 \Phi_4) + (t \partial_8 + 3H) \Phi_4 = V \Phi_{12} \end{aligned}$$

$$\begin{aligned}
(13) \quad & -\partial_4\Phi_2 + u(\partial_1\Phi_4 - \partial_2\Phi_3 + \partial_3\Phi_2) \\
& + v(\partial_5\Phi_3 + \partial_6\Phi_4 + \partial_7\Phi_5) + (t\partial_8 + 3H)\Phi_5 = V\Phi_{13} \\
(14) \quad & \partial_4\Phi_1 + u(\partial_1\Phi_3 + \partial_2\Phi_4 - \partial_3\Phi_1) \\
& + v(\partial_5\Phi_4 - \partial_6\Phi_3 + \partial_7\Phi_6) + (t\partial_8 + 3H)\Phi_6 = V\Phi_{14} \\
(15) \quad & \partial_4\Phi_4 + u(-\partial_1\Phi_2 + \partial_2\Phi_1 + \partial_3\Phi_4) \\
& + v(-\partial_5\Phi_1 + \partial_6\Phi_2 + \partial_7\Phi_7) + (t\partial_8 + 3H)\Phi_7 = V\Phi_{15} \\
(16) \quad & -\partial_4\Phi_3 + u(-\partial_1\Phi_1 - \partial_2\Phi_2 - \partial_3\Phi_3) \\
& + v(-\partial_5\Phi_2 - \partial_6\Phi_1 + \partial_7\Phi_8) + (t\partial_8 + 3H)\Phi_8 = V\Phi_{16}
\end{aligned}$$

In every component the gravitational term $3H\Phi_B$ appears next to the hidden-direction derivative $\tan z\partial_8\Phi_B$ of the same component B ($\gamma^{(x_8)}$ is a permutation matrix with entries ± 1).

Exact solutions used as test points. They satisfy the field equation for every a_4 .

- $U = 0$: $\Phi = \sin^\alpha z (\cosh(kx_4) + \frac{\sinh(kx_4)}{k}M)\chi$ with constant χ (16 free complex constants), $M = -m\gamma^{(x_4)} + 3H(2\alpha + 1)\gamma^{(x_4)}\gamma^{(x_8)}$, $k^2 = 9H^2(2\alpha + 1)^2 - m^2$ and α arbitrary; $M^2 = k^2$ and $M^TC + CM = 0$. Honest scope: at these solutions the pressures and the momentum density vanish and $\rho = mS$ is constant in x_4 , so they test the terms $\tan z\gamma^{(x_8)}\partial_8$ and $3H\gamma^{(x_8)}$ but are a weak test of conservation; the general proof of conservation is the Noether identity of section 7.
- $U = \frac{\lambda}{2}S^2$, homogeneous (nonlinear): $\Phi = (\cosh(kx_4) + \frac{\sinh(kx_4)}{k}M)\chi$, $M = -(m + \lambda S_0)\gamma^{(x_4)} + 3H\gamma^{(x_4)}\gamma^{(x_8)}$, $S_0 = \chi^\dagger C\chi$, $k^2 = 9H^2 - (m + \lambda S_0)^2$: $S = S_0$ is constant, the field equation holds, $\nabla_\mu T^\mu_\nu = 0$ for all ν , and $\rho = mS_0 + \lambda S_0^2/2$, $p = \lambda S_0^2/2$.

statement	Wolfram checks (theory)	sympy checks (theory)
Euler-Lagrange equations derived from $\mathcal{L}$	EL_Psibar_C, EL_Psi_C	commuting_euler_lagrange_psibar_variation, commuting_euler_lagrange_psi_variation
the adjoint equation is the Dirac conjugate explicit form, 16 components, chiral blocks	adjoint_equation_is_Dirac_conjugate_C, Dirac_operator_explicit_C, block_form	commuting_adjoint_equation_is_conjugate_comparison_of_field_equation_components and field_equation_blocks (sympy record, section comparison_with_wolfram)
evolution form exact solutions	evolution_form_C, solution_matrix_square, exact_solution_x4_x8_C, exact_solution_nonlinear_homogeneous_C	exact_solution_family_x4_x8, exact_nonlinear_homogeneous_solution

6 Non-triviality [2]: theorem and proof

The author's requirement [2]: the Lagrangian of `dirac16complex00` must be non-trivial, i.e. its Euler-Lagrange equations always possess nonzero contributions from gravity through the canonical spin connection unless the spacetime is flat 4+4, and they must be self-consistent, checked and verified.

Theorem (NON-TRIVIALITY [2]). Hypotheses: the author's metric of section 1 with $H > 0$ and $0 < z < \pi/2$; a_4 any differentiable function of x_4 (constant included); the diagonal vielbein of section 3 and its canonical spin connection; Φ the commuting 16-component field with the Lagrangian of section 4 and any real potential U . Then statements (1) to (4) hold.

(1) The gravitational term of the field equation. The spin-connection part of the Euler-Lagrange operator is

$$\gamma^\mu D_\mu \Phi - \gamma^\mu \partial_\mu \Phi = \gamma^\mu \Omega_\mu \Phi, \quad \gamma^\mu \Omega_\mu = 3H \gamma^{(x_8)},$$

a constant matrix, independent of a_4 , x_4 and x_8 . The term $3H \gamma^{(x_8)} \Phi$ is nonzero for every $H > 0$ and every $\Phi \neq 0$, because $\gamma^{(x_8)}$ is invertible, $(\gamma^{(x_8)})^2 = 1$.

(2) What cancels and what survives. Per direction (no sum)

$$\begin{aligned} \gamma^{x_i} \Omega_{x_i} &= \frac{a'_4}{2} \gamma^{(x_4)} + \frac{H}{2} \gamma^{(x_8)} & (i = x_1, x_2, x_3), \\ \gamma^{x_t} \Omega_{x_t} &= -\frac{a'_4}{2} \gamma^{(x_4)} + \frac{H}{2} \gamma^{(x_8)} & (t = x_5, x_6, x_7), \end{aligned}$$

and $\gamma^{x_4} \Omega_{x_4} = \gamma^{x_8} \Omega_{x_8} = 0$. The time-direction ($\gamma^{(x_4)}$) terms of the 3 inflating and the 3 deflating directions cancel exactly, $3a'_4/2 - 3a'_4/2 = 0$; the hidden-direction ($\gamma^{(x_8)}$) terms of all six directions add, $6 \cdot H/2 = 3H$.

(3) It vanishes only in flat 4+4 space. $\Omega_\mu = 0$ for all μ if and only if $a'_4 = 0$ and $H = 0$; $H = 0$ is a degenerate limit outside the family, so for every admissible metric the spin connection is nonzero. No choice of frame removes it: $[D_\mu, D_\nu] = \frac{1}{2} R_{ab\mu\nu} S^{ab}$ and $R^{x_8}_{x_8} = -6H^2 \neq 0$.

(4) The vielbein in every derivative term. Gravity also enters every derivative term,

$$\gamma^\mu \partial_\mu = e^{-a_4} \sin^{-1/6} z \gamma^{(x_i)} \partial_i + \gamma^{(x_4)} \partial_4 + e^{a_4} \sin^{-1/6} z \gamma^{(x_t)} \partial_t + \tan z \gamma^{(x_8)} \partial_8$$

(sums over i and t), none of the factors being identically 1; a_4 enters the field equation only through these factors (its spin-connection terms cancel by (2)).

Proof. Step (a), the inflating directions $i = x_1, x_2, x_3$: by section 3

$$\gamma^{x_i} = e^{-a_4} \sin^{-1/6} z \gamma^{(x_i)}, \quad \Omega_{x_i} = \frac{1}{2} e^{a_4} \sin^{1/6} z (a'_4 \gamma^{(x_i)} \gamma^{(x_4)} + H \gamma^{(x_i)} \gamma^{(x_8)}).$$

The scale factors cancel and $(\gamma^{(x_i)})^2 = \eta^{ii} = +1$, so $\gamma^{x_i} \Omega_{x_i} = \frac{1}{2} (a'_4 \gamma^{(x_4)} + H \gamma^{(x_8)})$.

Step (b), the deflating extra times $t = x_5, x_6, x_7$:

$$\gamma^{x_t} = e^{a_4} \sin^{-1/6} z \gamma^{(x_t)}, \quad \Omega_{x_t} = -\frac{1}{2} e^{-a_4} \sin^{1/6} z (a'_4 \gamma^{(x_4)} \gamma^{(x_t)} + H \gamma^{(x_t)} \gamma^{(x_8)}).$$

With $\gamma^{(x_t)} \gamma^{(x_4)} \gamma^{(x_t)} = -\gamma^{(x_4)} (\gamma^{(x_t)})^2 = \gamma^{(x_4)}$ and $(\gamma^{(x_t)})^2 = -1$ this gives $\gamma^{x_t} \Omega_{x_t} = -\frac{1}{2} (a'_4 \gamma^{(x_4)} - H \gamma^{(x_8)})$.

Step (c): $\Omega_{x_4} = \Omega_{x_8} = 0$. Summing (a), (b) and (c) gives statements (1) and (2). Equivalently, since $\{\gamma^\mu, \Omega_\mu\} = 0$ for each μ ,

$$\gamma^\mu \Omega_\mu = \frac{1}{2\sqrt{|g|}} \partial_\mu (\sqrt{|g|} e_a^\mu) \gamma^{(a)};$$

only $\mu = x_8$ contributes: $\sqrt{|g|} e_{x_8}^{x_8} = \cos z \tan z = \sin z$ and $\frac{1}{2\cos z} \partial_8 \sin z = 3H$, while $\partial_4 (\sqrt{|g|} e_{x_4}^{x_4}) = \partial_4 \cos z = 0$.

Step (d): each of the 12 nonzero $\omega_{\mu ab}$ is a'_4 or H times $\pm e^{\pm a_4} \sin^{1/6} z \neq 0$, and the S^{ab} are linearly independent; this gives statement (3), with the curvature statement from the integrability identity and the Ricci components of section 3.

Step (e): statement (4) is the explicit form of section 5. Every step is verified exactly, for the author's T16 and, in the a_4 record, also for a second, independently built Clifford representation. $\square$

What survives and what cancels, in words. The three inflating directions contribute $+\frac{a'_4}{2} \gamma^{(x_4)}$ each and the three deflating extra times $-\frac{a'_4}{2} \gamma^{(x_4)}$ each: the expansion-rate terms cancel exactly because 3-space inflates at the rate at which the extra times deflate (the same balance that makes $\sqrt{|g|} = \cos z$ independent of x_4). The warp $\sin^{1/6} z$ of the six directions along the hidden direction gives $+\frac{H}{2} \gamma^{(x_8)}$ for each of them and survives as $3H \gamma^{(x_8)}$. In the quantum theory of `dirac16complex` the same term is what makes the hidden-direction mode operator Hermitian; for the classical `dirac16complex00` it is a term of the field equation that no frame or coordinate choice removes.

statement	Wolfram checks	sympy checks
(1) $\gamma^\mu \Omega_\mu = 3H \gamma^{(x_8)}$, nonzero for every $\Phi \neq 0$	<code>nontriviality_2_</code> <code>dirac16complex00,</code> <code>gammaOmega_equal</code> <code>s_3H_gamma_x8 (theory);</code> <code>gravity_term_gamma</code> <code>ma_mu_Omega_mu (a4 record)</code>	<code>gamma_mu_Omega_m</code> <code>u_equals_3H_gamma_x8</code> <code>(theory); ownrep_gravity_t</code> <code>erm, authorT16_gravit</code> <code>y_term (a4 record)</code>
(2) the x_4 terms cancel, the x_8 terms add	<code>gammaOmega_x4_te</code> <code>rms_cancel, gammaOme</code> <code>ga_divergence_form</code>	<code>time_terms_cance</code> <code>l_hidden_term_survives,</code> <code>divergence_of_sq</code> <code>rtg_gamma</code>
(3) $\Omega_\mu = 0$ only for $a'_4 = 0$ and $H = 0$; never flat	<code>Omega_vanishes_i</code> <code>ff_a4prime_and_H_vanish,</code> <code>never_flat_for_H</code> <code>_positive, spin_cur</code> <code>vature_equals_Riemann</code>	<code>nontriviality_0m</code> <code>ega_zero_iff_flat,</code> <code>curvature_nonzer</code> <code>o_flat_only_formally,</code> <code>spinor_curvature</code> <code>_equals_riemann</code>
(4) explicit operator	<code>Dirac_operator_e</code> <code>xplicit_C</code>	

statement	Wolfram checks	sympy checks
the same term in the quantum theory of dirac16complex	good_sector_herm iticity_curved	

7 Self-consistency of the field equations

All statements are exact and verified for the commuting field.

Reality and derivation. $\mathcal{L}$ is real; the Euler-Lagrange equations derived from it are the field equation and its Dirac conjugate (sections 4 and 5).

Integrability of the connection. For all 28 pairs,

$$[D_\mu, D_\nu]\Phi = (\partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu])\Phi = \frac{1}{2}R_{ab\mu\nu}S^{ab}\Phi.$$

Second-order consistency (Lichnerowicz identity). Identically for a general field,

$$(\gamma^\mu D_\mu)^2\Phi = g^{\mu\nu}(D_\mu D_\nu - \Gamma^\lambda_{\mu\nu}D_\lambda)\Phi - \frac{R}{4}\Phi, \quad R = 6((a'_4)^2 - 7H^2),$$

and $\gamma^\mu\gamma^\nu F_{\mu\nu} = -\frac{R}{2}$ for the spinor curvature $F_{\mu\nu} = [D_\mu, D_\nu]$.

Cauchy problem. The equation is first order in x_4 with non-characteristic slices (evolution form, section 5).

Conserved current. $J^\mu = -i\bar{\Phi}\gamma^\mu\Phi$ is real, and off shell

$$\partial_\mu(\sqrt{|g|}J^\mu) = -i\sqrt{|g|}(\bar{E}\Phi + \bar{\Phi}E),$$

with $E = \gamma^\mu D_\mu\Phi - (m + U'(S))\Phi$ and $\bar{E} = (D_\mu\bar{\Phi})\gamma^\mu + (m + U'(S))\bar{\Phi}$ (the convention of the Wolfram record), so $\nabla_\mu J^\mu = 0$ on shell. The charge density is $J^{x_4} = \Phi^\dagger B\Phi$, and the conserved charge $Q = \int \cos z \Phi^\dagger B\Phi d^7x$ is an indefinite form (B has signature (8,8)).

Bianchi-type identity of the energy-momentum tensor. The off-shell Noether identity of diffeomorphism invariance,

$$\partial_\mu(\sqrt{|g|}T^\mu{}_\lambda) - \sqrt{|g|}T^\mu{}_a\partial_\lambda e^a{}_\mu = \sum_\phi \text{EL}_\phi\partial_\lambda\phi \quad (\lambda = x_1, \dots, x_8),$$

and the off-shell identity of local Lorentz invariance (the antisymmetric part of T is a combination of the Euler-Lagrange expressions EL_ϕ) give: on shell T is symmetric and $\nabla_\mu T^\mu{}_\nu = 0$, exactly, for general solutions.

Gravity side. Each Lovelock tensor is divergence-free, $\nabla_\mu E_{(k)}^\mu{}_\nu = 0$, and the constraint is propagated by the evolution equation (section 11).

Test points. The exact solutions of section 5 satisfy the field equation, current conservation and $\nabla_\mu T^\mu{}_\nu = 0$ exactly.

statement	Wolfram checks (theory)	sympy checks (theory)
integrability $[D_\mu, D_\nu] = \frac{1}{2} R_{ab\mu\nu} S^{ab}$ Lichnerowicz identity	spin_curvature_e quals_Riemann Lichnerowicz_identity_C	spinor_curvature _equals_riemann lichnerowicz_id entity_on_fields, lichnerowicz_con traction
current: reality, conservation, indefinite charge	current_real_C, current_ conservation_identity_C, charge_density_i s_Krein_form_C	commuting_curren t_conservation
Noether identities and conservation of T	Noether_identity _diffeomorphisms_C, Noether_identity _local_Lorentz_C, conservation_on_ shell_general	commuting_empt_co nservation_on_shell, commuting_empt_co nservation_negat ive_control

8 The energy-momentum tensor (classical bilinears)

For `dirac16complex00` the energy-momentum tensor is the classical bilinear (a c-number function of Φ ; no operator ordering, no normal ordering).

Definition and closed form. $T^\nu{}_\mu = e^b{}_\mu \frac{1}{\sqrt{|g|}} \delta S_{\text{act}} / \delta e^b{}_\nu$ (exact first-order variation of $\sqrt{|g|}$, γ^μ and the spin connection; S_{act} the action) equals, for all 64 components,

$$T^\nu{}_\mu = \delta^\nu{}_\mu L_0 - \frac{1}{2} (\bar{\Phi} \gamma^\nu D_\mu \Phi - D_\mu \bar{\Phi} \gamma^\nu \Phi) - \frac{1}{4} g_{\mu\rho} \nabla_\lambda (\bar{\Phi} \{\gamma^\lambda, \Sigma^{\nu\rho}\} \Phi), \quad \Sigma^{\nu\rho} = \frac{1}{4} [\gamma^\nu, \gamma^\rho],$$

with $L_0 = \mathcal{L} / \sqrt{|g|}$; the last term (the totally antisymmetric spin density) comes from the variation of the spin connection. Its symmetric part is the Belinfante tensor

$$T^\nu{}_\mu = \delta^\nu{}_\mu L_0 - \frac{1}{4} (\bar{\Phi} \gamma^\nu D_\mu \Phi - D_\mu \bar{\Phi} \gamma^\nu \Phi + \bar{\Phi} \gamma_\mu D^\nu \Phi - D^\nu \bar{\Phi} \gamma_\mu \Phi),$$

and on shell the variation tensor is symmetric and equals it (section 7; independently, with all 64 components of a first-order vielbein variation around the author's metric, the variation tensor equals the Belinfante tensor on shell for 64 of 64 components). The normalisation is $\delta S_{\text{act}} = \frac{1}{2} \int \sqrt{|g|} T^{\mu\nu} \delta g_{\mu\nu}$.

Sign convention. The energy density is $\rho = -T^{x_4}{}_{x_4}$ and the pressures are $p_\mu = T^\mu{}_\mu$ (no sum) for the space-like and the extra-time directions:

$$\rho = -T^{x_4}{}_{x_4}, \quad p_3 = T^{x_1}{}_{x_1}, \quad p_t = T^{x_5}{}_{x_5}, \quad p_8 = T^{x_8}{}_{x_8}.$$

Diagonal components in the primordial field. With the kinetic term of direction μ (no sum)

$$K_\mu = \frac{1}{2f_\mu} (\bar{\Phi} \gamma^{(\mu)} \partial_\mu \Phi - \partial_\mu \bar{\Phi} \gamma^{(\mu)} \Phi),$$

that is

$$\begin{aligned} K_{x_i} &= \frac{1}{2} e^{-a_4} \sin^{-1/6} z (\bar{\Phi} \gamma^{(x_i)} \partial_i \Phi - \partial_i \bar{\Phi} \gamma^{(x_i)} \Phi), & K_{x_4} &= \frac{1}{2} (\bar{\Phi} \gamma^{(x_4)} \partial_4 \Phi - \partial_4 \bar{\Phi} \gamma^{(x_4)} \Phi), \\ K_{x_t} &= \frac{1}{2} e^{a_4} \sin^{-1/6} z (\bar{\Phi} \gamma^{(x_t)} \partial_t \Phi - \partial_t \bar{\Phi} \gamma^{(x_t)} \Phi), & K_{x_8} &= \frac{1}{2} \tan z (\bar{\Phi} \gamma^{(x_8)} \partial_8 \Phi - \partial_8 \bar{\Phi} \gamma^{(x_8)} \Phi), \end{aligned}$$

one has

$$L_0 = \sum_\mu K_\mu - mS - U(S), \quad T^\mu{}_\mu = L_0 - K_\mu \quad (\text{no sum}).$$

No spin-connection term survives in the diagonal components ($\{\gamma^\mu, \Omega_\mu\} = 0$), and the variation tensor and the Belinfante tensor agree on the diagonal off shell. Hence, for every configuration (off shell),

$$\begin{aligned} \rho &= - \sum_{\mu \neq x_4} K_\mu + mS + U(S), \\ p_3 &= \sum_{\mu \neq x_1} K_\mu - mS - U(S) \quad (= T^{x_2}{}_{x_2} = T^{x_3}{}_{x_3} \text{ for states isotropic in 3-space}), \\ p_t &= \sum_{\mu \neq x_5} K_\mu - mS - U(S), & p_8 &= \sum_{\mu \neq x_8} K_\mu - mS - U(S). \end{aligned}$$

The x_4 - x_8 component (the off-diagonal component allowed by the symmetry of the metric; $\Omega_{x_4} = \Omega_{x_8} = 0$):

$$\begin{aligned} T^{x_4}{}_{x_8} &= -\frac{1}{4} (B_{48} - \cot z B_{84}), & T^{x_8}{}_{x_4} &= -\tan^2 z T^{x_4}{}_{x_8}, \\ B_{48} &= \bar{\Phi} \gamma^{(x_4)} \partial_8 \Phi - \partial_8 \bar{\Phi} \gamma^{(x_4)} \Phi, & B_{84} &= \bar{\Phi} \gamma^{(x_8)} \partial_4 \Phi - \partial_4 \bar{\Phi} \gamma^{(x_8)} \Phi; \end{aligned}$$

it vanishes on homogeneous on-shell states.

Trace. Off shell $T^\mu{}_\mu = 8L_0 - \sum_\mu K_\mu = 7 \sum_\mu K_\mu - 8(mS + U)$. Exactly, $\sum_\mu K_\mu = (m + U')S + \frac{1}{2}(\bar{\Phi} E - \bar{E} \Phi)$, so on shell $\sum_\mu K_\mu = (m + U'(S))S$, $L_0 = S U'(S) - U(S)$ and

$$T^\mu{}_\mu = -mS + 7S U'(S) - 8U(S) = -mS + 3\lambda S^2 \quad (U = \frac{\lambda}{2} S^2).$$

Conservation and energy exchange. For a diagonal homogeneous tensor

$$T = \text{diag}(p_3, p_3, p_3, -\rho, p_t, p_t, p_t, p_8)(x_4)$$

one has $\nabla_\mu T^\mu{}_{x_4} = -\rho' - 3a'_4(p_3 - p_t)$ and $\nabla_\mu T^\mu{}_{x_8} = 3H \cot z (2p_8 - p_3 - p_t)$, so conservation gives

$$\frac{d\rho}{dx_4} = -3 a'_4 (p_3 - p_t), \quad p_3 + p_t = 2p_8 :$$

energy flows between 3-space and the extra times unless $p_3 = p_t$, since $\sqrt{|g|}$ does not depend on x_4 .

statement	Wolfram checks (theory)	sympy checks (theory)
variation tensor in closed form	T_vielbein_variation_closed_form_C	commuting_empt_eq uals_general_vielbein_variation_on_shell
Belinfante tensor, symmetric	T_symmetric_part_Belinfante_C	commuting_empt_symmetric
normalisation $\delta S_{\text{act}} = \frac{1}{2} \int \sqrt{ g } T^{\mu\nu} \delta g_{\mu\nu}$, diagonal components x_4 - x_8 component	T_diagonal_components_C T_x4_x8_component_C	commuting_empt_eq uals_vielbein_variation_diagonal commuting_T_x4x8_homogeneous
trace, on-shell kinetic sum	EMT_trace_C, kinetic_sum_on_shell_C	commuting_trace_on_shell
energy exchange and the x_8 condition	energy_exchange_equation	energy_exchange_equation

9 Kinetic energy, potential energy, energy density and pressures

Split. $T^\nu{}_\mu = T_{\text{kin}}{}^\nu{}_\mu + T_{\text{pot}}{}^\nu{}_\mu$ with the potential part and the kinetic (derivative) part

$$T_{\text{pot}}{}^\nu{}_\mu = -\delta^\nu{}_\mu (mS + U(S)),$$

$$T_{\text{kin}}{}^\nu{}_\mu = \delta^\nu{}_\mu \sum_\lambda K_\lambda - \frac{1}{4} (\bar{\Phi} \gamma^\nu D_\mu \Phi - D_\mu \bar{\Phi} \gamma^\nu \Phi + \bar{\Phi} \gamma_\mu D^\nu \Phi - D^\nu \bar{\Phi} \gamma_\mu \Phi)$$

(formula EMT_kinetic_potential of Revision/theory/field-theory.json, compared with the sympy split).

Densities (per unit proper 7-volume; the energy in a region of a slice $x_4 = \text{const}$ is the integral with the weight $\sqrt{|g|} = \cos z$):

quantity	kinetic part	potential part	total
energy density $\rho = -T^{x_4}{}_{x_4}$	$-\sum_{\mu \neq x_4} K_\mu$	$mS + U(S)$	$-\sum_{\mu \neq x_4} K_\mu + mS + U$
3-space pressure $p_3 = T^{x_1}{}_{x_1}$	$\sum_{\mu \neq x_1} K_\mu$	$-(mS + U)$	$\sum_{\mu \neq x_1} K_\mu - mS - U$
extra-time pressure $p_t = T^{x_5}{}_{x_5}$	$\sum_{\mu \neq x_5} K_\mu$	$-(mS + U)$	$\sum_{\mu \neq x_5} K_\mu - mS - U$
hidden-direction pressure $p_8 = T^{x_8}{}_{x_8}$	$\sum_{\mu \neq x_8} K_\mu$	$-(mS + U)$	$\sum_{\mu \neq x_8} K_\mu - mS - U$

Homogeneous states on shell (Φ depending on x_4 only): $K_\mu = 0$ for $\mu \neq x_4$ and $K_{x_4} = (m + U'(S))S$. So the kinetic energy density vanishes, $-T_{\text{kin}}^{x_4}_{x_4} = 0$, the kinetic pressure is $(m + U'(S))S$ in every direction, and

$$\rho = mS + U(S), \quad p_3 = p_t = p_8 = S U'(S) - U(S);$$

for $U = \frac{\lambda}{2}S^2$: $\rho = mS + \frac{\lambda}{2}S^2$ and $p = \frac{\lambda}{2}S^2$ (formula `EMT_homogeneous_on_shell` of `Revision/theory/field-theory.json`; the exact nonlinear solution of section 5 realises these values). S is constant along x_4 for a homogeneous solution, so ρ and p are constant, consistent with $d\rho/dx_4 = -3a'_4(p_3 - p_t) = 0$.

Sign of the energy. $S = \Phi^\dagger C \Phi$ is an indefinite form (section 2), and for the homogeneous solutions of section 5 $S = S_0 = \chi^\dagger C \chi$ with an arbitrary constant χ ; the sign of $\rho = mS + U$ is therefore not fixed by the field equations. This record contains no positivity statement for the energy of `dirac16complex00`.

statement	Wolfram checks	sympy checks
homogeneous on-shell ρ , p ; kinetic and potential parts	<code>exact_solution_n</code> <code>online_homogeneous_C</code> (theory)	<code>commuting_homogeneous_on_shell_rho_p</code> (theory)
S constant along x_4 for a homogeneous solution	<code>condensate_S_constant</code> (a_4 record)	<code>ownrep_condensate_e_S_constant</code> , <code>authorT16_condensate_S_constant</code> (a_4 record)

10 Equations of state

From the diagonal components (SPEC section 4):

$$w_3 = \frac{p_3}{\rho}, \quad w_t = \frac{p_t}{\rho}, \quad w_8 = \frac{p_8}{\rho}.$$

For homogeneous states on shell the three are equal,

$$w_3 = w_t = w_8 = \frac{S U'(S) - U(S)}{mS + U(S)}, \quad w = \frac{\lambda S}{2m + \lambda S} \quad (U = \frac{\lambda}{2}S^2)$$

For $\lambda = 0$ the homogeneous state has $p = 0$ and $w = 0$. Because S , ρ and p are constant for a homogeneous solution, these ratios are constant in x_4 .

statement	source
definitions and homogeneous values of w_3 , w_t , w_8	formula <code>equation_of_state_definitions</code> of <code>Revision/theory/field-theory.json</code> ; sympy check <code>commuting_homogeneous_on_shell_rho_p</code>

Not computed in this document: the equation of state that a 3-space observer infers (after integrating over the hidden direction and the extra times), its time dependence as the extra times deflate, CPL parameters, and any comparison with supernova data. These belong to the dark-sector investigation of Hypothesis00 (SPEC section 8), which is a separate Revision record.

11 The field equations for a4[x4]

Einstein-Lovelock equations (SPEC section 5):

$$\sum_{k=1}^3 \alpha_k E_{(k)}{}^\mu{}_\nu + \Lambda \delta^\mu{}_\nu = \kappa T^\mu{}_\nu, \quad E_{(k)} = -\frac{P_{(k)}}{2^{k+1}}, \quad E_{(1)} = G,$$

with $P_{(k)}{}^h{}_j$ the Lovelock tensors computed with the generalized Kronecker delta (GKD) from the Riemann tensor (MTW convention). $P_{(4)}$ vanishes in 8 dimensions (a GKD with 9 indices over 8 values), so $k = 1, 2, 3$ is the complete series. The components computed directly from the curvature equal the GKD record `Revision/gkd_lovelock/results/lovelock-tensors.json` monomial by monomial. The mixed Riemann components contain neither the warp nor e^{a_4} , so the left-hand sides depend on H , a'_4 and a''_4 only.

The Lovelock components. All off-diagonal components vanish (including x_4 - x_8); $x_2 = x_3 = x_1$ and $x_6 = x_7 = x_5$:

$$\begin{aligned} E_{(1)}{}^{x_1}{}_{x_1} &= -3(a'_4)^2 + a''_4 + 15H^2, & E_{(1)}{}^{x_4}{}_{x_4} &= 3(a'_4)^2 + 21H^2, \\ E_{(1)}{}^{x_5}{}_{x_5} &= -3(a'_4)^2 - a''_4 + 15H^2, & E_{(1)}{}^{x_8}{}_{x_8} &= 15H^2 - 3(a'_4)^2, \\ E_{(2)}{}^{x_1}{}_{x_1} &= 12(a'_4)^4 - 24(a'_4)^2 a''_4 + 168(a'_4)^2 H^2 - 40a''_4 H^2 - 180H^4, \\ E_{(2)}{}^{x_4}{}_{x_4} &= -36(a'_4)^4 - 120(a'_4)^2 H^2 - 420H^4, \\ E_{(2)}{}^{x_5}{}_{x_5} &= 12(a'_4)^4 + 24(a'_4)^2 a''_4 + 168(a'_4)^2 H^2 + 40a''_4 H^2 - 180H^4, \\ E_{(2)}{}^{x_8}{}_{x_8} &= 12(a'_4)^4 + 168(a'_4)^2 H^2 - 180H^4, \\ \\ E_{(3)}{}^{x_1}{}_{x_1} &= -72(a'_4)^6 + 360(a'_4)^4 a''_4 - 648(a'_4)^4 H^2 + 432(a'_4)^2 a''_4 H^2 \\ &\quad - 1944(a'_4)^2 H^4 + 360a''_4 H^4 + 360H^6, \\ E_{(3)}{}^{x_4}{}_{x_4} &= 360(a'_4)^6 + 648(a'_4)^4 H^2 + 1080(a'_4)^2 H^4 + 2520H^6, \\ E_{(3)}{}^{x_5}{}_{x_5} &= -72(a'_4)^6 - 360(a'_4)^4 a''_4 - 648(a'_4)^4 H^2 - 432(a'_4)^2 a''_4 H^2 \\ &\quad - 1944(a'_4)^2 H^4 - 360a''_4 H^4 + 360H^6, \\ E_{(3)}{}^{x_8}{}_{x_8} &= -72(a'_4)^6 - 648(a'_4)^4 H^2 - 1944(a'_4)^2 H^4 + 360H^6. \end{aligned}$$

Each $E_{(k)}$ is divergence-free, and $\sum_h P_{(k)}{}^h{}_h = (8 - 2k)L_{(k)}$.

The independent equations for a general source $T = \text{diag}(p_3, p_3, p_3, -\rho, p_t, p_t, p_t, p_8)$ plus $q_{48} = T^{x_4}{}_{x_8}$ and $q_{84} = T^{x_8}{}_{x_4}$:

- constraint (x_4 , first order in a_4): $\sum_k \alpha_k E_{(k)}{}^{x_4}{}_{x_4} + \Lambda = -\kappa\rho$;

- 3-space ($x_1 = x_2 = x_3$): $\sum_k \alpha_k E_{(k)}^{x_1}_{x_1} + \Lambda = \kappa p_3$;
- extra times ($x_5 = x_6 = x_7$): $\sum_k \alpha_k E_{(k)}^{x_5}_{x_5} + \Lambda = \kappa p_t$;
- hidden direction (x_8 , first order in a_4): $\sum_k \alpha_k E_{(k)}^{x_8}_{x_8} + \Lambda = \kappa p_8$;
- off-diagonal: $0 = \kappa q_{48} = \kappa q_{84}$, and $0 = \kappa T^\mu_\nu$ for every other $\mu \neq \nu$.

The evolution equation (3-space minus extra time) contains a_4'' and is sourced by $p_3 - p_t$:

$$a_4'' F(a_4') = \kappa (p_3 - p_t), \quad F = 720(a_4')^4 \alpha_3 - 48(a_4')^2 \alpha_2 + 864(a_4')^2 \alpha_3 H^2 + 2\alpha_1 - 80\alpha_2 H^2 + 720\alpha_3 H^4;$$

F vanishes identically only for $\alpha_1 = \alpha_2 = \alpha_3 = 0$.

Consistency conditions.

- Algebraic: $\sum_k \alpha_k (E^{x_1}_{x_1} + E^{x_5}_{x_5} - 2E^{x_8}_{x_8}) = 0$ identically, so the equations force $p_3 + p_t = 2p_8$.
- x_8 -dependence: the left-hand sides are free of x_8 (and of a_4 itself), so every component of the source must be a function of x_4 alone and $q_{48} = q_{84} = 0$; an x_8 -dependent source is not compatible with the metric of section 1.
- Isotropy: since the left-hand sides of x_1, x_2, x_3 coincide (and those of x_5, x_6, x_7), the source must have $T^{x_1}_{x_1} = T^{x_2}_{x_2} = T^{x_3}_{x_3}$ and $T^{x_5}_{x_5} = T^{x_6}_{x_6} = T^{x_7}_{x_7}$.
- Conservation: $\nabla_\mu T^\mu_{x_4} = -\partial_4 \rho - 3a_4'(p_3 - p_t) + \partial_8 q_{84} - 6H \tan z q_{84}$ and $\nabla_\mu T^\mu_{x_8} = \partial_8 p_8 + 3H \cot z (2p_8 - p_3 - p_t) + \partial_4 q_{48}$; with an x_8 -independent source and $q_{48} = q_{84} = 0$ they reduce to $\rho' = -3a_4'(p_3 - p_t)$ and $p_3 + p_t = 2p_8$, both implied by the field equations (Bianchi identity).
- Constraint propagation: $\frac{d}{dx_4} \sum_k \alpha_k E_{(k)}^{x_4}_{x_4} = 3a_4' \sum_k \alpha_k (E^{x_1}_{x_1} - E^{x_5}_{x_5})$ identically, so the derivative of the constraint is $3a_4'$ times the evolution equation once $\rho' = -3a_4'(p_3 - p_t)$ holds; for $a_4' \neq 0$ the evolution equation follows from the constraint and conservation.

Einstein gravity ($\alpha_1 = 1, \alpha_2 = \alpha_3 = 0$):

$$\begin{aligned} 3(a_4')^2 + 21H^2 + \Lambda &= -\kappa\rho, & -3(a_4')^2 + a_4'' + 15H^2 + \Lambda &= \kappa p_3, \\ -3(a_4')^2 - a_4'' + 15H^2 + \Lambda &= \kappa p_t, & -3(a_4')^2 + 15H^2 + \Lambda &= \kappa p_8, \\ 2a_4'' &= \kappa(p_3 - p_t), & \kappa(p_8 + \rho) &= -6(a_4')^2 - 6H^2. \end{aligned}$$

Every C^2 function a_4 is allowed with the source it requires; that source always has $\kappa(\rho + p_8) = -6((a_4')^2 + H^2) < 0$ (for $\kappa > 0$ it violates the null energy condition along $x_4 + x_8$), and $T = 0$ has no solution for $H > 0$ and any Λ .

The linear member $a_4 = AHx_4 + a_0$ ($a_4' = AH, a_4'' = 0$; 3-space scale factor e^{AHx_4} , extra-time scale factor e^{-AHx_4} : exponential deflation of the extra times for $A > 0$). For any α_k the field equations require $p_3 = p_t = p_8 = p$ and ρ , all constant, with

$$\begin{aligned}
\rho &= -\frac{360A^6\alpha_3H^6}{\kappa} + \frac{36A^4\alpha_2H^4}{\kappa} - \frac{648A^4\alpha_3H^6}{\kappa} - \frac{3A^2\alpha_1H^2}{\kappa} + \frac{120A^2\alpha_2H^4}{\kappa} \\
&\quad - \frac{1080A^2\alpha_3H^6}{\kappa} - \frac{21\alpha_1H^2}{\kappa} + \frac{420\alpha_2H^4}{\kappa} - \frac{2520\alpha_3H^6}{\kappa} - \frac{\Lambda}{\kappa}, \\
p &= -\frac{72A^6\alpha_3H^6}{\kappa} + \frac{12A^4\alpha_2H^4}{\kappa} - \frac{648A^4\alpha_3H^6}{\kappa} - \frac{3A^2\alpha_1H^2}{\kappa} + \frac{168A^2\alpha_2H^4}{\kappa} \\
&\quad - \frac{1944A^2\alpha_3H^6}{\kappa} + \frac{15\alpha_1H^2}{\kappa} - \frac{180\alpha_2H^4}{\kappa} + \frac{360\alpha_3H^6}{\kappa} + \frac{\Lambda}{\kappa};
\end{aligned}$$

a vacuum ($\rho = p = 0$) needs

$$\mathcal{V} = 72A^4\alpha_3H^4 - 8A^2\alpha_2H^2 + 144A^2\alpha_3H^4 + \alpha_1 - 40\alpha_2H^2 + 360\alpha_3H^4 = 0,$$

which in Einstein gravity ($\mathcal{V} = 1$) is impossible. The equations contain A only through A^2 : the deflating branch $A > 0$ (a_4 increasing) and the branch $A < 0$ satisfy the same equations.

statement	Wolfram checks (a_4 record)	sympy checks (a_4 record)
metric, volume, Riemann free of warp and e^{a_4}	metric_is_SPEC_section_1, sqrt_abs_det_g_is_cos_z, mixed_riemann_free_of_warp_and_a4	mixed_riemann_free_of_warp_and_a4, riemann_entries_laurent
$P_{(k)}$ from the curvature equal the GKD record; traces; $P_{(4)} = 0$	P1_direct_equals_minus_4_Einstein, P1_direct_equals_gkd_branch_monomials, P2_direct_equals_gkd_branch_monomials, P3_direct_equals_gkd_branch_monomials, P1_trace_identity, P2_trace_identity, P3_trace_identity, P4_vanishes_pigeonhole	P1_equals_minus_4_Einstein, P1_equal_s_gkd_branch_monomials, P2_equals_gkd_branch_monomials, P3_equal_s_gkd_branch_monomials
divergence-free $E_{(k)}$	E1_divergence_free, E2_divergence_free, E3_divergence_free	E1_divergence_free, E2_divergence_free, E3_divergence_free
independent components, structure	independent_components	other_components_vanish, json_lovelock_components, json_general_source_equations
evolution equation and F	evolution_factorises_a4pp_times_F, evolution_F_not_identically_zero	evolution_factorises, evolution_F_not_identically_zero
algebraic condition, first-order constraint and x_8 equation	algebraic_identity_x1_plus_x5_minus_2x8, x8_component_contains_no_a4pp	algebraic_identity, constraint_and_x8_first_order

statement	Wolfram checks (a_4 record)	sympy checks (a_4 record)
conservation, constraint propagation	conservation_components, constraint_propagation_bianchi	conservation_components, bianchi_x4
Einstein case	einstein_components, einstein_null_energy_x8, einstein_no_vacuum_solution	einstein_components, einstein_null_energy_x8, einstein_no_vacuum, json_einstein
linear member	linear_member_eq, ual_pressures, linear_member_vacuum_factor	linear_member_eq, ual_pressures, linear_member_vacuum_factor, json_linear_member

12 dirac16complex00 as the source of the a4 equations

The tensor. The source is the classical bilinear tensor of section 8. The a_4 record writes it as $T^\mu{}_\nu = \sigma_T(-K^\mu{}_\nu + \delta^\mu_\nu L_0)$ with $K^\mu{}_\nu = \frac{1}{2}(\bar{\Phi}\gamma^\mu D_\nu\Phi - D_\nu\bar{\Phi}\gamma^\mu\Phi)$ symmetrised, and $\sigma_T = +1$; this is the Belinfante tensor of section 8 (whose symmetrised kinetic term is the same expression), so the two records use the same tensor and the same sign. With $k_\mu = K^\mu{}_\mu$ (no sum; equal to K_μ of section 8) and $L_0 = \sum_\mu k_\mu - mS - U(S)$:

$$\rho = k_4 - L_0, \quad p_3 = L_0 - k_1, \quad p_t = L_0 - k_5, \quad p_8 = L_0 - k_8, \quad q_{48} = -K^{(x_4)}_{x_8}.$$

The equations with dirac16complex00 as the source (classical commuting Φ ; k_μ , S and U classical bilinears):

$$\begin{aligned} \sum_k \alpha_k E_{(k)}^{x_4} x_4 + \Lambda &= -\kappa(k_4 - L_0), & \sum_k \alpha_k E_{(k)}^{x_8} x_8 + \Lambda &= \kappa(L_0 - k_8), \\ a_4'' F(a_4') &= \kappa(k_5 - k_1), & 2k_8 &= k_1 + k_5, & K^{(\mu)}{}_\nu &= 0 \quad (\mu \neq \nu), \end{aligned}$$

together with $k_1 = k_2 = k_3$, $k_5 = k_6 = k_7$ and the requirement that every component be independent of x_8 . On shell $L_0 = S U'(S) - U(S)$. The evolution equation shows what drives a_4' : a difference between the kinetic terms along 3-space and along the extra times. This record contains no exact non-homogeneous dirac16complex00 configuration that satisfies all of these conditions with $k_1 \neq k_5$; whether one exists is open.

The homogeneous condensate. For $\Phi = \Phi(x_4)$ on shell the field equation is $\partial_4\Phi = A_\Phi\Phi$ with $A_\Phi = -\gamma^{(x_4)}(M - 3H\gamma^{(x_8)})$ and $M = m + U'(S)$, free of x_8 . Proved exactly (Wolfram, and in the sympy record both for an independently built Clifford representation and for the author's T16):

- $CA_\Phi + A_\Phi^T C = 0$, so $S = \bar{\Phi}\Phi$ is constant along x_4 ;
- the diagonal kinetic components are $K^{x_4}{}_{x_4} = MS$ and 0 for every other direction, and $K^{x_4}{}_{x_8} = K^{x_8}{}_{x_4} = 0$; hence $\rho = mS + U(S)$, $p_3 = p_t = p_8 = S U'(S) - U(S)$ and $q_{48} = q_{84} = 0$, and the algebraic condition and the x_8 -independence hold;

- the nonzero off-diagonal kinetic components are exactly multiples of the 15 three-gamma bilinears $B_{abc} = \bar{\Phi}\gamma^{(a)}\gamma^{(b)}\gamma^{(c)}\Phi$ with $\{a, b, c\} = \{i, x_4, x_8\}$ ($i = x_1..x_3, x_5..x_7$) and $\{i, j, x_4\}$ ($i = x_1..x_3, j = x_5..x_7$); there are 42 nonzero ordered pairs (μ, ν) .

The 42 components fall into ten groups; within each group the coefficient is the same for every 3-space index x_i ($i = 1, 2, 3$) and extra-time index x_t ($t = 5, 6, 7$), and on the patch $\sqrt[12]{\cot^2 z + 1} = \sin^{-1/6} z$ (the gammas in this table are the frame gammas; generated from `Revision/field_equations_a4/a4-equations.json`):

component	bilinear	coefficient
$K^{x_i}_{x_4}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_8}\Phi$	$\frac{7}{4}e^{-a_4}\sqrt[12]{\cot^2 z + 1}H$
$K^{x_i}_{x_t}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_t}\Phi$	$-\frac{1}{2}e^{-2a_4}(a'_4)$
$K^{x_i}_{x_8}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_8}\Phi$	$\frac{1}{4}e^{-a_4}(a'_4)\cot z\sqrt[12]{\cot^2 z + 1}$
$K^{x_4}_{x_i}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_8}\Phi$	$-\frac{7e^{a_4}H}{4\sqrt[12]{\cot^2 z + 1}}$
$K^{x_4}_{x_t}$	$\bar{\Phi}\gamma^{x_4}\gamma^{x_t}\gamma^{x_8}\Phi$	$-\frac{7e^{-a_4}H}{4\sqrt[12]{\cot^2 z + 1}}$
$K^{x_t}_{x_i}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_t}\Phi$	$\frac{1}{2}e^{2a_4}(a'_4)$
$K^{x_t}_{x_4}$	$\bar{\Phi}\gamma^{x_4}\gamma^{x_t}\gamma^{x_8}\Phi$	$-\frac{7}{4}e^{a_4}\sqrt[12]{\cot^2 z + 1}H$
$K^{x_t}_{x_8}$	$\bar{\Phi}\gamma^{x_4}\gamma^{x_t}\gamma^{x_8}\Phi$	$\frac{1}{4}e^{a_4}(a'_4)\cot z\sqrt[12]{\cot^2 z + 1}$
$K^{x_8}_{x_i}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_8}\Phi$	$\frac{e^{a_4}(a'_4)}{4\cot z\sqrt[12]{\cot^2 z + 1}}$
$K^{x_8}_{x_t}$	$\bar{\Phi}\gamma^{x_4}\gamma^{x_t}\gamma^{x_8}\Phi$	$-\frac{e^{-a_4}(a'_4)}{4\cot z\sqrt[12]{\cot^2 z + 1}}$

The off-diagonal field equations $0 = \kappa T^\mu_\nu$ therefore require $B_{ix_4x_8} = 0$ for the six i (coefficients proportional to H and to a'_4) and, when $a'_4 \neq 0$, $B_{ijx_4} = 0$ for the nine pairs i, j .

Exact witnesses that these conditions can be met with $S \neq 0$: $\Phi = e^{-i\omega x_4}\Phi_0$ with $\omega = \sqrt{M^2 - 9H^2}$ ($M^2 > 9H^2$), and $\Phi_0 = v_1 + c v_2$ built from the two eigenvectors of A_Φ in the sectors $\gamma^{(x_1)}\gamma^{(x_5)} = \gamma^{(x_2)}\gamma^{(x_6)} = \gamma^{(x_3)}\gamma^{(x_7)} = -1$ and $+1$, $c = v_1^\dagger C v_2$. At $(M, H) = (5, 1)$, $(5, 4/3)$ and $(-5, 1)$ (exact Gaussian-rational arithmetic) the frequencies are $\omega = 4, 3, 4$, all 15 three-gamma bilinears and every off-diagonal kinetic component vanish for every a_4 and a'_4 , and $S = 204800, 115200, 204800$ respectively.

Theorem (the condensate allows only the linear member). Hypotheses: Φ a homogeneous on-shell condensate as above (classical, commuting) satisfying the off-diagonal conditions; $a_4 \in C^2$; $(\alpha_1, \alpha_2, \alpha_3) \neq (0, 0, 0)$. Then $p_3 = p_t$, so $a''_4 F(a'_4) = 0$; F is a nonzero polynomial in a'_4 with finitely many roots and a'_4 is continuous, so $a''_4 = 0$ on every interval:

$$a_4 = AHx_4 + a_0,$$

and the deflating member (for $A > 0$) is the only kind of a_4 this source allows (`theoremLinear` of `Revision/field_equations_a4/a4-equations.json`, resting on the checks listed below).

The source it requires. With $a'_4 = AH$ the remaining equations are the constraint and the x_8 equation, at $a'_4 = AH$:

$$\kappa(mS + U) = -\left(\Lambda + \sum_k \alpha_k E_{(k)}^{x_4 x_4}\right), \quad \kappa(SU' - U) = \Lambda + \sum_k \alpha_k E_{(k)}^{x_8 x_8},$$

that is, $\rho = mS + U$ and $p = SU' - U$ must equal the linear-member values of section 11. In Einstein gravity

$$\kappa(mS + U) = -(3A^2 + 21)H^2 - \Lambda, \quad \kappa(SU' - U) = (15 - 3A^2)H^2 + \Lambda,$$

and for $U = \frac{\lambda}{2}S^2$ and $m \neq 0$ these are equivalent to

$$\kappa mS = -(36H^2 + 2\Lambda), \quad 6(A^2 + 1)H^2 = -\kappa S(m + \lambda S),$$

so $A^2 = -\kappa S(m + \lambda S)/(6H^2) - 1$, and a real A needs $\kappa S(m + \lambda S) \leq -6H^2$. For the condensate $m + \lambda S = M$ is the effective mass of the field equation and $S(m + \lambda S) = \rho + p$; the requirement is the null-energy statement of section 11 for this source. S takes both signs (section 2), so the sign condition is not excluded by the algebra; the three exact witnesses above have $S > 0$ with $M = 5, 5, -5$. This record contains no check that combines a witness with the Einstein conditions (definite κ , Λ , m , λ and A): that the deflating member is realised by a specific `dirac16complex00` condensate is not established here.

statement	Wolfram checks (a_4 record)	sympy checks (a_4 record)
the Clifford facts used, in a second representation	<code>C_properties_own_rep,</code> <code>gamma_mu_anticom</code> <code>mutates_with_Omega</code> <code>_mu_no_sum</code>	<code>ownrep_clifford,</code> <code>ownrep_C_properties,</code> <code>ownrep_anticommu</code> <code>tator_no_sum, authorT1</code> <code>6_clifford, authorT1</code> <code>6_C_properties, authorT1</code> <code>6_anticommutator_no_sum</code>
condensate equation free of x_8 ; S constant; adjoint	<code>condensate_equat</code> <code>ion_x8_consistent,</code> <code>condensate_S_constant,</code> <code>condensate_adjoi</code> <code>nt_equation</code>	<code>ownrep_condensat</code> <code>e_S_constant, ownrep_c</code> <code>ondensate_adjoint,</code> <code>authorT16_conden</code> <code>sate_S_constant,</code> <code>authorT16_conden</code> <code>sate_adjoint</code>
diagonal kinetic tensor of the condensate	<code>condensate_kinet</code> <code>ic_tensor_diagonal</code>	<code>ownrep_condensat</code> <code>e_kinetic_diagonal,</code> <code>authorT16_conden</code> <code>sate_kinetic_diagonal</code>
off-diagonal components: the 15 three-gamma bilinears, coefficients	<code>condensate_offdi</code> <code>agonal_are_three</code> <code>_gamma_bilinears</code>	<code>ownrep_condensat</code> <code>e_offdiagonal_th</code> <code>ree_gamma, authorT1</code> <code>6_condensate_off</code> <code>diagonal_three_gamma,</code> <code>json_offdiagonal</code> <code>_coefficients, offdiago</code> <code>nal_coefficients</code> <code>_representation</code> <code>independent</code>
exact witnesses	<code>condensate_diago</code> <code>nal_witness_exact</code>	<code>ownrep_condensat</code> <code>e_witness, authorT1</code> <code>6_condensate_witness</code>

statement	Wolfram checks (a_4 record)	sympy checks (a_4 record)
the linear member is the only one ($F \neq 0$)	evolution_F_not_identically_zero, linear_member_eq ual_pressures	evolution_F_not_identically_zero, linear_member_eq ual_pressures
Einstein conditions with $U = \frac{\lambda}{2}S^2$	condensate_einst ein_quadratic_U	condensate_einst ein_quadratic_U

13 Pairing theorems for dirac16complex00 (summary)

The pairing of universes of masses $\{+m, -m\}$ is the subject of `Revision/docs/PAIR_CREATION_PROOFS`; for the commuting field the Revision pairing records prove exactly the following.

T1 (chirality pairing). Hypotheses: any gravitational field, fixed and the same for both members (a test field; the metric is not varied); $\Gamma = \text{diag}(-I_8, I_8)$; $U = \frac{\lambda}{2}S^2$, or any function U with $U \rightarrow -U$. Then, identically,

$$\mathcal{L}_{m,\lambda}[\Gamma\Phi] = -\mathcal{L}_{-m,-\lambda}[\Phi], \quad S[\Gamma\Phi] = S[\Phi], \quad T^{(-m,-\lambda)}[\Gamma\Phi] = -T^{(m,\lambda)}[\Phi], \quad J[\Gamma\Phi] = -J[\Phi]$$

(for a general U : $\mathcal{L}_{m,U}[\Gamma\Phi] = -\mathcal{L}_{-m,-U}[\Phi]$), and Φ solves the (m, λ) equations if and only if $\Gamma\Phi$ solves the $(-m, -\lambda)$ ones. The pair's total energy-momentum density, current and charge vanish as classical c-number bilinears.

T2 (mirror pairing). With the isometry $\phi : x_8 \rightarrow \pi/(6H) - x_8$ ($z \rightarrow \pi - z$, the Z_2 mirror across $z = \pi/2$, an ASSUMED construction at a degenerate surface of the metric) and $\Phi'(\phi(x)) = \gamma^{(x_8)}\Phi(x)$:

$$\mathcal{L}_{m,\lambda}[\Phi'](\phi(x)) = \mathcal{L}_{-m,\lambda}[\Phi](x), \quad S' = -S,$$

and the mirror partner carries the same energy density and charge (the tensor transformed by the reflection), not the opposite.

There is no quantum reading for dirac16complex00: it is not quantised. These theorems map solutions to solutions; they derive no creation process, rate or amplitude (the lists of what is not established in the pairing records).

statement	Wolfram checks (pairing)	sympy checks (pairing)
T1 for the commuting field in the primordial metric	Gamma_properties, T1_kernel_field_ equation, T1_linea rity_argument, T1_Lagra ngian_primordial _commuting, T1_Lagra ngian_negative_c ontrols_primordi al_commuting, T1_field _equation_covari ance_primordial_ commuting, T1_Euler _Lagrange_map_pr imordial_commuting, T1_energy_moment um_primordial_commuting, T1_current_primo rdial_commuting, T1_general_poten tial_primordial_ commuting	T1.metric.commut ing.lagrangian, T1.metri c.commuting.nega tive_controls, T1.metri c.commuting.eule r_lagrange_map, T1.metri c.commuting.emt, T1.metric.commut ing.pair_total_emt_zero, T1.metric.commut ing.current, T1.metri c.commuting.gene ral_potential
T2 for the commuting field	T2_mirror_is_isometry, T2_mirror_Lagran gian_commuting, T2_mirro r_energy_momentu m_and_current_commuting, T2_mirror_Euler_ Lagrange_commuting	T2.metric.commut ing.euler_lagrange_map, T2.metric.commuting.emt, T2.metric.commut ing.current, T2.metri c.commuting.S_odd

14 Check index

Every check below has the verdict PASS in the named report (the publication test re-reads the reports and confirms each name and verdict). Counts at the time of writing: wolfram-algebra 45 checks, python-algebra 35, wolfram-field-theory 84, python-field-theory 70, wolfram-a4-report 47, python-a4-report 61, wolfram-pairing 99, python-pairing 64; all pass.

report	checks cited in this document
Revision/algebra/reports/wolfram-algebra.json	Clifford_relation, reality, symmetry _pattern, eta_in_author_order, coordinate_map, C_real_symmetric
Revision/algebra/reports/wolfram-algebra.json	C_squared_identity, C_gamma_real_ant isymmetric, Gamma_diag, B_Hermitian, B_squared_identity, B_signature_8_8
Revision/algebra/reports/wolfram-algebra.json	S_Lorentz_algebra, S_preserves_C, S_commutates_with_Gamma, Pin44_ir reducible_commutant_dim_1, Spin44_c ommutant_dim_2_chiral_projectors, chiral_halves_irreducible

report	checks cited in this document
Revision/algebra/reports/wolfram-algebra.json	chiral_halves_inequivalent_inter twiners_0, Spin_Pin_reflection_swap s_halves
Revision/algebra/reports/python-algebra.json	clifford_relation, clifford_relatio n_sympy, reality_signed_permutations, symmetry_pattern, C_equals_noteboo k_sigma16, C_real_symmetric_involut ion
Revision/algebra/reports/python-algebra.json	C_gamma_antisymmetric, chiralit y_diag, chirality_anticommutates, B_hermitian_involution_signature, S_lorentz_algebra, S_preserves_C_an d_commutates_with_Gamma
Revision/algebra/reports/python-algebra.json	clifford_products_span_M16, pin_comm utant_dimension_1, even_products_sp an_M8_plus_M8, spin_commutant_d imension_2, spin_halves_irreducible, spin_halves_inequivalent
Revision/algebra/reports/python-algebra.json	reflections_exchange_halves, fixture_comparison_gammas_json
Revision/theory/reports/wolfram-field-theory.json	metric_is_the_authors, vielbein _inverse, sqrt_det_g_is_cos_z, degenerate_at_H_0, christoffel_count, ricci_scalar
Revision/theory/reports/wolfram-field-theory.json	ricci_mixed_components, never_fl at_for_H_positive, vielbein_postula te, omega_antisymmetric, omega_co mponents, Omega_components
Revision/theory/reports/wolfram-field-theory.json	gamma_covariantly_constant, spin_cur vature_equals_Riemann, gammaOme ga_equals_3H_gamma_x8, gammaOme ga_x4_terms_cancel, gammaOmega_diver gence_form, L_real_C
Revision/theory/reports/wolfram-field-theory.json	L_total_divergence_to_unsymmetri sed_C, L_spin_connection_drops_out_C, EL_Psibar_C, EL_Psi_C, adjoint_ equation_is_Dirac_conjugate_C, Dirac_operator_explicit_C
Revision/theory/reports/wolfram-field-theory.json	block_form, evolution_form_C, nontriviality_2_dirac16complex00, Omega_vanishes_iff_a4prime_and_H _vanish, current_real_C, current_ conservation_identity_C
Revision/theory/reports/wolfram-field-theory.json	charge_density_is_Krein_form_C, Lichnerowicz_identity_C, Majorana _Lg_total_derivative_grassmann, Majorana_Lg_commuting_control, T_vielbein_variation_closed_form_C, T_symmetric_part_Belinfante_C

report	checks cited in this document
Revision/theory/reports/wolfram-field-theory.json	Noether_identity_diffeomorphisms_C, Noether_identity_local_Lorentz_C, conservation_on_shell_general, T_diagonal_components_C, EMT_trace_C, kinetic_sum_on_shell_C
Revision/theory/reports/wolfram-field-theory.json	T_x4_x8_component_C, energy_exchange_equation, solution_matrix_square, exact_solution_x4_x8_C, exact_solution_nonlinear_homogeneous_C, good_sector_hermiticity_curved
Revision/theory/reports/python-field-theory.json	gammas_json_equals_python_construction, metric_from_vielbein_equals_SPEC, sqrt_det_g_equals_cos_z, christoffel_symmetric_metric_compatible, spin_connection_antisymmetric, vielbein_postulate
Revision/theory/reports/python-field-theory.json	Omega_x4_and_Omega_x8_vanish, covariant_constancy_D_mu_gamma_nu, gamma_mu_Omega_mu_equals_3H_gamma_x8, time_terms_cancel_hidden_term_survives, anticommutator_gamma_Omega_vanishes, divergence_of_sqrtg_gamma
Revision/theory/reports/python-field-theory.json	nontriviality_Omega_zero_iff_flat, curvature_nonzero_flat_only_formally, spinor_curvature_equals_riemann, lichnerowicz_contraction, lichnerowicz_identity_on_fields, commuting_lagrangian_real
Revision/theory/reports/python-field-theory.json	commuting_controls_not_vacuous, commuting_total_divergence_relation, commuting_euler_lagrange_psi_variation, commuting_euler_lagrange_psi_variation, commuting_adjoint_equation_is_conjugate, commuting_current_conservation
Revision/theory/reports/python-field-theory.json	commuting_empt_symmetric, commuting_empt_conservation_on_shell, commuting_empt_conservation_negative_control, commuting_trace_on_shell, commuting_homogeneous_on_shell_rho_p, commuting_T_x4x8_homogeneous

report	checks cited in this document
Revision/theory/reports/python-field-theory.json	commuting_emt_equals_general_vielbein_variation_on_shell, commuting_emt_equals_vielbein_variation_diagonal, negative_control_majorana_grassmann_total_derivative, negative_control_majorana_commuting_contrast, energy_exchange_equation, exact_solution_family_x4_x8
Revision/theory/reports/python-field-theory.json	exact_nonlinear_homogeneous_solution
Revision/field_equations_a4/reports/wolfram-a4-report.json	metric_is_SPEC_section_1, sqrt_abs_det_g_is_cos_z, mixed_riemann_free_of_warp_and_a4, P1_direct_equals_minus_4_Einstein, P1_direct_equals_gkd_branch_monomials, P2_direct_equals_gkd_branch_monomials, P3_direct_equals_gkd_branch_monomials, P1_trace_identity, P2_trace_identity, P3_trace_identity, P4_vanishes_pigeonhole, E1_divergence_free
Revision/field_equations_a4/reports/wolfram-a4-report.json	E2_divergence_free, E3_divergence_free, independent_components, evolution_factorises_a4pp_times_F, evolution_F_not_identically_zero, algebraic_identity_x1_plus_x5_minus_2x8
Revision/field_equations_a4/reports/wolfram-a4-report.json	x8_component_contains_no_a4pp, constraint_propagation_bianchi, conservation_components, einstein_components, einstein_null_energy_x8, einstein_no_vacuum_solution
Revision/field_equations_a4/reports/wolfram-a4-report.json	linear_member_equal_pressures, linear_member_vacuum_factor, C_properties_own_rep, gamma_mu_anticommutates_with_Omega_mu_no_sum, gravity_term_gamma_mu_Omega_mu, condensate_equation_x8_consistent
Revision/field_equations_a4/reports/wolfram-a4-report.json	condensate_S_constant, condensate_adjoint_equation, condensate_kinetic_tensor_diagonal, condensate_offdiagonal_are_three_gamma_bilinears, condensate_einstein_quadratic_U, condensate_diagonal_withn_ess_exact
Revision/field_equations_a4/reports/python-a4-report.json	mixed_riemann_free_of_warp_and_a4, riemann_entries_laurent, P1_equals_minus_4_Einstein, P1_equals_gkd_branch_monomials, P2_equals_gkd_branch_monomials, P3_equals_gkd_branch_monomials

report	checks cited in this document
Revision/field_equations_a4/reports/python-a4-report.json	E1_divergence_free, E2_divergence_free, E3_divergence_free, evolution_factorises, evolution_F_not_identically_zero, algebraic_identity_constraint_and_x8_first_order, bianchi_x4, other_components_vanish, conservation_components, einstein_components, einstein_null_energy_x8_einstein_no_vacuum, linear_member_equal_pressures, linear_member_vacuum_factor, condensate_einstein_quadratic_U, json_lovelock_components, json_general_source_equations
Revision/field_equations_a4/reports/python-a4-report.json	json_einstein, json_linear_member, ownrep_clifford, ownrep_C_properties, ownrep_anticommutator_no_sum, ownrep_gravity_term
Revision/field_equations_a4/reports/python-a4-report.json	ownrep_condensate_S_constant, ownrep_condensate_adjoint, ownrep_condensate_kinetic_diagonal, ownrep_condensate_offdiagonal_three_gamma, ownrep_condensate_witness, authorT16_clifford
Revision/field_equations_a4/reports/python-a4-report.json	authorT16_C_properties, authorT16_anticommutator_no_sum, authorT16_gravity_term, authorT16_condensate_S_constant, authorT16_condensate_adjoint, authorT16_condensate_kinetic_diagonal
Revision/field_equations_a4/reports/python-a4-report.json	authorT16_condensate_offdiagonal_three_gamma, authorT16_condensate_witness, json_offdiagonal_coefficients, offdiagonal_coefficients_representation_independent
Revision/pairing/reports/wolfram-pairing.json	Gamma_properties, T1_kernel_field_equation, T1_linearity_argument, T1_Lagrangian_primordial_commuting, T1_Lagrangian_negative_controls_primordial_commuting, T1_field_equation_covariance_primordial_commuting
Revision/pairing/reports/wolfram-pairing.json	T1_Euler_Lagrange_map_primordial_commuting, T1_energy_momentum_primordial_commuting, T1_current_primordial_commuting, T1_general_potential_primordial_commuting, T2_mirror_is_isometry, T2_mirror_Lagrangian_commuting
Revision/pairing/reports/wolfram-pairing.json	T2_mirror_energy_momentum_and_current_commuting, T2_mirror_Euler_Lagrange_commuting

report	checks cited in this document
Revision/pairing/reports/python-pairing.json	T1.metric.commuting.lagrangian, T1.metric.commuting.negative_controls, T1.metric.commuting.euler_lagrange_map, T1.metric.commuting.emt, T1.metric.commuting.pair_total_emt_zero, T1.metric.commuting.current
Revision/pairing/reports/python-pairing.json	T1.metric.commuting.general_potential, T2.metric.commuting.euler_lagrange_map, T2.metric.commuting.emt, T2.metric.commuting.current, T2.metric.commuting.S_odd

15 What is not claimed

- **No quantisation.** `dirac16complex00` is not quantised; its energy-momentum tensor is a classical bilinear. The operator statements (Krein space, Fock space, normal ordering) belong to `dirac16complex`.
- **No positivity of the energy.** $S = \Phi^\dagger C \Phi$ is indefinite; no statement that the energy of `dirac16complex00` is bounded below is made or derived.
- **No 3-space observer equation of state, no CPL fit, no comparison with supernova data.** Section 10 gives the ratios p/ρ of the 8-dimensional tensor only; Hypothesis00 (a time-varying dark-energy or dark-matter equation of state from `dirac16complex00`) is investigated in the separate dark-sector record, not here. For the homogeneous condensate the ratio w is constant in x_4 .
- **No general solution.** The exact solutions of section 5 are test points; the field equation is not solved in general, and no existence statement for the nonlinear Cauchy problem is made beyond the first-order evolution form.
- **No realisation of the deflating member by a specific condensate.** The theorem of section 12 shows that a homogeneous condensate allows only $a_4 = AHx_4 + a_0$ and gives the conditions the source must satisfy; it is not shown that a specific `dirac16complex00` condensate meets those conditions with definite κ , Λ , m , λ and $A > 0$, and no non-homogeneous source with $k_1 \neq k_5$ (which would drive a_4'') is constructed.
- **The x_8 dependence.** The metric of section 1 is an exact solution only with an x_8 -independent, isotropic source with vanishing off-diagonal components; a `dirac16complex00` configuration that depends on x_8 is not a consistent source for this metric.
- **The brane.** The patch end $z = \pi/2$ is a degenerate surface of the metric ($g_{88} = 0$, $\sqrt{|g|} = 0$); no junction condition is derived, and the mirror construction used by T2 is an assumption.
- **Pairing.** The pairing theorems of section 13 derive no creation process, rate or amplitude for universes of masses $\{+m, -m\}$; they are correspondences between solutions, in a fixed gravitational field.

- **Notebook conventions.** The canonical spin connection used here is the one with $\omega_{\mu ab} = \eta_{ac}\omega_{\mu}{}^c{}_b$; the notebook’s contraction of the mixed components (which lacks a metric factor) is not used, and its Majorana-type L_g serves only as the negative control of section 4.

16 Reproduction

From the repository root, in this order (each verifier writes its report and exits with code 0 only if every check passes; two runs give byte-identical outputs):

```
wolframscript -file Revision/algebra/wolfram/verify_algebra.wls
python Revision/algebra/python/check_algebra.py
wolframscript -file Revision/theory/wolfram/verify_field_theory.wls
python Revision/theory/python/check_field_theory.py
wolframscript -file Revision/field_equations_a4/wolfram/verify_field_equations_a4.wls
python Revision/field_equations_a4/python/check_field_equations_a4.py
wolframscript -file Revision/pairing/wolfram/verify_pairing.wls
python Revision/pairing/python/check_pairing.py
```

Re-run for this document (2026-10-01). The eight verifiers were run in this order on a scratch copy of the Revision inputs (the committed files were not touched). Run times on the development machine: algebra 4 s (Wolfram) and 2 s (sympy); theory 1029 s (Wolfram) and 115 s (sympy); a_4 equations 15 s and 4 s; pairing 82 s and 147 s. Every run exited with code 0 and every check passed. All outputs are byte-identical to the committed files except the sympy theory report `Revision/theory/reports/python-field-theory.json`: its 70 checks are identical, and it differs only in the section `comparison_with_wolfram`. The committed report was compared with an earlier Wolfram run (82 checks; the current Wolfram report has 84). Compared with the current Wolfram output, 63 of 63 check pairs and 26 of 27 formula records agree; the one record reported as disagreeing, `energy_exchange`, fails because the comparison parses an older text form of that formula (an equation written with `==`), whereas the current `Revision/theory/field-theory.json` writes it as `nabla_mu T^mu_x4 = -rho'[x4] - 3 a4'[x4] (p3 - p_t) = 0`, i.e. $d\rho/dx_4 = -3 a_4'(x_4) (p_3 - p_t)$. That is the same equation as section 8 of this document, and the check `energy_exchange_equation` verifies it on both sides; the parser mismatch is left to the owner of `Revision/theory`.

The document is built and checked with (the first command on one line)

```
python scripts/build_provenance_pdf.py Revision/docs/DIRAC16COMPLEX00_FIELD_THEORY.md
--developer-layout --specifications Revision/pdf-specifications.json
python -m unittest Revision/tests/test_dirac16complex00_field_theory_publication.py -v
```

The builder makes the .tex twice and the PDF in two fresh directories, requires byte identity and a warning-free LaTeX log, and verifies the PDF against its entry in Revision’s own registry `Revision/pdf-specifications.json`. The publication test checks that the .tex is the builder’s output for this .md, that the PDF is registered with its page count and sha256, that the formula blocks of sections 5, 11 and 12 are the ones generated from `Revision/theory/field-theory.json` and `Revision/field_equations_a4/a4-equations.json`, and that every check cited in section 14 exists in its report with a passing verdict.